

BRTS

BRTS India | Features, Types, Examples , Advantages & Disadvantages

29/10/2020 / By NOSPlan / Transportation

Introduction

The UN report of 2014 states that about 54 percent of the world's population resides in urban areas. This count is assumed to rise to a whopping 66 percent by the year 2050. This could pose an issue for the public transportation system in cities where they are already operating at breaking point. Migrants often rely on public transportation for their daily commute which in turn increases congestion. Public transport system congestion has become a major problem in cities like New York , Mumbai etc., which has to be addressed immediately. The increasing need for urban mass transit mobility is being addressed in the cities all across the world by following the best practices in the world. BRTS is one of the many practices followed.

What is BRTS?

Bus Rapid Transit System or commonly known as BRTS is a high-quality transit system which is based on buses. Its features are mostly similar to that of metro rails. BRTS consists of one single lane which can transport up to 20,000 passengers in one direction per hour. It is one of the most successful transport systems, which reduces delays, accidents, congestion and various other problems. This system is highly accepted in India because it helps the society in achieving their needs. The construction of BRTS is quite simple when compared to other transit systems. Big budgets or high-level technologies are not required as the existing roads itself can be converted into BRTS lanes.

History of BRTS

BRTS in World

The BRTS concept was implemented for the first time in Curitiba, Brazil in 1974. It was developed by the then mayor, Jaime Lerner and was called 'Surface Metro'. This brilliant transportation system inspired various other countries to adopt similar systems in their cities. During the 1970s, development of BRTS was done mostly in North and South America, only after 1990s did the system make its way to other continents. Quito, Ecuador (1996), Los Angeles, USA (1999) and Bogotá, Columbia (2000) are few cities which have implemented this transit system. Bogotá's success drew attention of the world. It is the most famous and advanced BRTS in the world.

BRTS in India

In India, cities like Jaipur, Ahmadabad, Indore, Vijayawada, Delhi, Pune, Bhopal and Rajkot have successfully implemented BRTS, covering a total network of 120km. The total investment for these transit systems have been more than 13.6 billion rupees. The first BRTS in the country to be operated was Rainbow Bus Rapid Transport



Delhi BRTS



Rajkot BRTS



Surat BRTS



Pune BRTS



Indore BRTS



Ahmedabad BRTS

Why BRTS?

With the increase in population in urban areas, there is an equal amount of increase in demand for cheap, convenient and safe public transportation system in India. Rail-based transit systems are expensive to construct and maintain. The construction of rail-based systems take a long time, hence they cannot be in reach for a long time to come. People and the government prefer a new cost effective, fast mode of transport for the convenience and conveyance. BRTS full fills these requirement's, as it is fast, cost-effective, safe, and comfortable. BRTS can serve to all types of environment be it sub urban or urban. The buses can be driven by humans or electronically guided, can be driven on aerial structures, freeway medians and underground.

Features of BRTS

The features of BRTS are:

- **Separated lanes:** In BRTS system, the inner most lanes are allotted to buses, to ensure that congestion do not delay the buses.
- **Off board fare collection:** Passengers pay the bus fare in the station instead of paying on board. This reduces the passengers to cause delays in normal city buses.
- **Platform level boarding:** At grade bus lanes are preferred for easy access, operational flexibility and reducing costs.
- **Stations:** The station which is the entry point vary in design from simple stops with shelters to multi-modal terminals.
- Route maps, signage and real-time information are displayed in stations or in vehicles.

What is the structure of the BRT system?

A BRT system would consist of most of the following:

1. Station with off-board ticket collection
2. Alignment in the centre of the roads
3. Level of station platforms equal to the bus floor

Kickstart your IAS preparation now and complement it with the links given below:

- **IAS Questions & Answers for UPSC 2021-22**
- **Topic-wise IAS Prelims Questions with Solutions**
- **IAS Mock Tests**
- **Daily Video Analysis – The Hindu Newspaper**
- **UPSC Previous Year Question Papers**
- **Daily Press Information Bureau (PIB) Analysis**

Benefits of BRT

The benefits of the Bus Rapid Transit System are as follows:

- BRT reduces the overall amount of vehicle kilometres travelled in a city by shifting commuters to high-capacity buses that can carry up to 160 passengers at a time.
- Setting up a new BRT system also provides cities with an opportunity to scrap older, more polluting traditional vehicles.
- Introducing a new BRT corridor, therefore, has major implications not just for GHG

Benefits of BRT

The benefits of the Bus Rapid Transit System are as follows:

- BRT reduces the overall amount of vehicle kilometres travelled in a city by shifting commuters to high-capacity buses that can carry up to 160 passengers at a time.
- Setting up a new BRT system also provides cities with an opportunity to scrap older, more polluting traditional vehicles.
- Introducing a new BRT corridor, therefore, has major implications not just for GHG emissions, but also for air pollution.
- Implementing BRT systems contributes to reductions in traffic accidents and fatalities in several key ways.
- Dedicated bus lanes that separate BRT buses from mixed traffic, pre-paid boarding and level platforms speed up passenger boarding, whilst traffic signal management prioritizes BRT buses. High-frequency bus service also minimizes waiting times to help save travel time for passengers.

Name the First Bus rapid transit system in the world and India.

In 1974, the first bus rapid transit system was launched in Curitiba, Brazil. It was called the Rede Integrada de Transporte ('Integrated Transportation Network').

The first BRTS in India is Rainbow Bus Rapid Transit System in Pune, started in 2006.

What are the advantages of the Bus Rapid Transit Corridor?

The major advantages of the Bus Rapid Transit corridor are listed below:

What are the advantages of the Bus Rapid Transit Corridor?

The major advantages of the Bus Rapid Transit corridor are listed below:

1. The transit system has undergone major changes in terms of leveraging level boarding mechanisms that would eliminate the need for lifts or ramps.
2. This system helps in managing traffic efficiently and prevents bus bunching, which helps to keep the operations cost under track.
3. This system would help enhance the safety of people due to effective lighting arrangements at various places in and around the stations.
4. This system would be a life-saver in cases where lanes are required to be kept free from the passage for ambulances and fire brigade vehicles.
5. This would be an ultimate time-saver for daily passengers in cities with heavy traffic.

Why BRTS ?????

Problems with bus system

- Slow
- Unreliable
- Not always frequent
- Takes the long way around
- Uncomfortable
- Technologically backward
- No universal design

Gap
filling

Benefits of metro system

- Very fast
- Regular and frequent
- Travels long distance at less time
- comfortable
- High technology
- universal design
- Very high capacity

BRTS tries to solve issues of bus system and incorporates benefits of metro system

Advantages and Disadvantages of BRTS

Advantages	Disadvantages
Grade separated provides high sense of safety and saves travel time	Construction of graded lanes has high capital value
BRTS lanes do not interfere with mixed traffic	Lanes with no interference may not be attainable always
Designated curb side bus-only lanes improve BRT travel time making buses competitive with automobile	Implementation of new bus lanes with street widening may displace parking and pedestrian paths
Converted High Occupancy Vehicle (HOV) lanes, allow buses to operate faster, more reliably and more safely than buses in mixed traffic lanes	Since buses share lane with HOV's, automobiles may impede bus operations. This make HOV lanes less efficient than bus only lanes.

BRTS Models/Types

Level of BRTS	Characteristics
Full BRTS	• Metro quality service • Integrate network of routes and corridors • Closed, high quality stations • Off-board fare collection • Frequent and rapid service • Modern, clean vehicle • Marketing identity • Superior customer service
BRTS	• Segregated bus way • Typically, pre board fare payment • Higher quality stations • Clean vehicle technology • Marketing identity

BRTS Lite	• Some form of bus priority but not fully segregated bus ways • Improved travel time • Higher shelters • Clean vehicle technology • Marketing identity
Basic	• Segregated bus ways/ single corridor services
Busway	• On-board fare collection • Basic bus shelter

Conclusion

BRTS has proven itself to be successful and a sustainable mode of transportation throughout the world. It is over the edge when compared to other modes of transportation, given its high sense of safety through provision of separate lanes for the buses. The bus fare, quality of the buses, air conditioning adds up to its charm thus making it popular among the masses. The total construction expenditure is also significantly lower when compared to MRTS, since existing infrastructure is but to use. However, the implementation of BRTS has few challenges too, like, the city must have adequate number of cross sections in order to provide right of way to the buses, maintain the quality of mixed traffic lanes and minimize noise levels and

Metro

Metro rail is fast becoming the most popular means of mass rapid transit in India with an operational network of over **835 km** across **15 cities**. But did you know that the first metro in the country started operations all the way back in 1984 in Kolkata. It is also the only metro rail network that is controlled by the [Indian Railways](#); all others are operated by autonomous local authorities.



The second oldest after Kolkata Metro is Delhi Metro, which makes the largest and busiest network of all. Following the success of Delhi Metro, other cities in India also started exploring the option to implement metro projects. With this idea, more than INR 1 trillion was invested in metro projects in the year 2018-19.

History of Metro Rail in India

The introduction of metro rail in India began in Kolkata (formerly Calcutta) with the Kolkata Metro. It was the first underground metro system in the country and started its operations in 1984. Since then, metro networks have expanded to other major cities across India, including Delhi, Mumbai, Chennai, Bengaluru, Hyderabad, Jaipur, Kochi, and Lucknow.

Various government agencies and public-private partnerships have played significant roles in the development of metro networks in India. The primary organizations involved include:

- **Kolkata Metro:** The Kolkata Metro was the first metro network in India and was implemented by the Kolkata Metro Rail Corporation (KMRC).
- **Delhi Metro:** The Delhi Metro Rail Corporation (DMRC) played a pivotal role in planning, constructing, and operating the Delhi Metro. The DMRC is a joint venture between the Government of India and the Government of the National Capital Territory of Delhi.
- **Mumbai Metro:** The Mumbai Metro Rail Corporation Limited (MMRCL) is responsible for implementing metro projects in Mumbai. It is a joint venture between the Government of Maharashtra and the Government of India.
- **Chennai Metro:** The Chennai Metro Rail Limited (CMRL) is responsible for the implementation of the Chennai Metro. It is a joint venture between the Government of India and the Government of Tamil Nadu.

Who Invented Metro in India?

The development of the metro system in India can be attributed to multiple entities and organizations. The first modern metro system in India was the Kolkata Metro, which was the result of collaboration between Indian and foreign entities. The project was executed by the Kolkata Metro Rail Corporation (KMRC) and the planning and design assistance was provided by the Soviet Union (now Russia). The initial section of the Kolkata Metro was inaugurated on October 24, 1984.

Subsequently, other cities in India also started developing their metro systems. Delhi Metro, one of the most extensive and successful metro networks in India, was established by the Delhi Metro Rail Corporation (DMRC) under the leadership of E. Sreedharan, a renowned Indian engineer. E. Sreedharan, often referred to as the "Metro Man," played a crucial role in the planning, execution, and development of the Delhi Metro.

Underground Metro in India

Several cities in India have underground metro systems. Some notable examples include:

- **Kolkata Metro:** The Kolkata Metro is predominantly an underground metro system with both elevated and underground sections. It has been instrumental in easing transportation in the city. The idea of an underground metro in Kolkata was first proposed by Bidhan Chandra Roy, the then Chief Minister of West Bengal.
- **Delhi Metro:** The Delhi Metro consists of both elevated and underground sections. The underground portions are crucial for serving densely populated areas of the city.
- **Chennai Metro:** The Chennai Metro also features underground sections, providing efficient transportation options for the city's residents.

Besides, as many as **15 other cities in India have over 500 km of metro lines under different stages** of implementation. These include

- **Ahmedabad** – Gujarat Metro Rail Corporation Limited
- **Pune** – Pune Metropolitan Region Development Authority
- **Indore** – Madhya Pradesh Metro Rail Corporation
- **Bhopal** – Madhya Pradesh Metro Rail Corporation
- **Varanasi** – Uttar Pradesh Metro Rail Corporation
- **Kozhikode** – Kerala Rapid Transit Corporation Ltd
- **Vijayawada** – Amaravati Metro Rail Corporation (AMRC)
- **Meerut** – Uttar Pradesh Metro Rail Corporation
- **Visakhapatnam** – Amaravati Metro Rail Corporation (AMRC)
- **Kanpur** – Lucknow Metro Rail Corporation
- **Agra** – Uttar Pradesh Metro Rail Corporation
- **Coimbatore** – Chennai Metro Rail Limited
- **Patna** – Delhi Metro Rail Corporation
- **Navi Mumbai** – City and Industrial Development Corporation (CIDCO)
- **Guwahati** – Guwahati Metropolitan Development Authority